“Version control and project management using Git, GitHub, and the Waterfall Model.”

Git & GitHub in Long-Term Projects (6+ Months)

- **Git** is a **version control system** used to track code changes.
- **GitHub** is a **cloud platform** where Git repositories are stored and shared.
- Developers use GitHub to **collaborate, manage code versions, and review changes**.
- In long-term projects (6 months or more), Git helps track **daily progress and code history**.
- Teams use **branches** to work on different features without affecting the main project.
- After completing features, code is merged using **pull requests**.

Software Development Process (Waterfall Model)

The **Waterfall Model** follows a **step-by-step sequential process**.

1️⃣ Business Analysis / Requirement Analysis

- Understand **client requirements**.
- Gather project needs and objectives.
- Create **requirement documents**.
- Define project scope.

2️⃣ System Design

Design how the system will work.

Two main types of design:

High-Level Design (HLD)

- System architecture
- Modules and components
- Technology selection

Low-Level Design (LLD)

- Detailed module design
- Database structure
- Algorithms and logic

3️⃣ Implementation (Coding)

- Developers write code based on the design.
- Code is stored and managed using **Git repositories**.
- Features are developed using **branches** in GitHub.

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4 Testing

- Test the application to find bugs.
- Types of testing:
  - Unit Testing
  - Integration Testing
  - System Testing
  - User Acceptance Testing (UAT)

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5 Deployment

- The application is released to production.
- DevOps tools help automate deployment.
- Code from GitHub is deployed to servers or cloud platforms.

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6 Maintenance (Important Missing Step)

- Fix bugs after deployment.
- Update features.
- Improve system performance.
- Provide long-term support.